Inference-Time Acoustic Style Transfer in F5-TTS via Mel Conditioning Interpolation and Flow-Matching Initialisation Biasing

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Code: https://github.com/yardencarmel/SigProcForSpeech

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Abstract

F5-TTS [1] is a state-of-the-art zero-shot text-to-speech system based on Conditional Flow Matching (CFM) and Diffusion Transformers. Despite its impressive voice-cloning capability, the paper explicitly identifies the absence of fine-grained paralinguistic control - particularly emotion - as a key limitation. We reproduce the paper's sway-sampling and classifier-free guidance ablations, confirming that $s{=}-0.8$, NFE$=32$ achieves $0\,\%$ word error rate. We then introduce two complementary inference-time extensions that require no retraining. **Extension 1** injects a linearly interpolated blend of two reference mel spectrograms directly into the model's conditioning branch, producing a smooth acoustic gradient between speakers measurable via speaker similarity and mel-cepstral distortion. **Extension 2** adapts the SDEdit (stochastic differential edit) framework [4] to flow-matching TTS by biasing the ODE starting point toward a style reference while conditioning on an identity reference; four variants are evaluated: SDEdit initialisation (A), style guidance via a two-pass ODE (B), scheduled conditioning switching (C), and noise statistics transfer (D). We discuss why extending F5-TTS to unseen languages such as Hebrew requires full retraining, and analyse a novel security concern: the noise initialisation channel, once exposed as a controllable interface, constitutes a potential adversarial injection surface.

Keywords: text-to-speech; acoustic style transfer; flow matching; SDEdit; mel spectrogram; paralinguistic control; adversarial examples.

1 Introduction

Zero-shot voice cloning - generating speech in an arbitrary speaker's voice from a short reference clip without speaker-specific training - is a central challenge in modern text-to-speech (TTS). Early neural TTS systems such as Tacotron 2 [14] and FastSpeech 2 [15] required speaker-specific fine-tuning and could not generalise to unseen voices. Autoregressive (AR) models like VALL-E [11] addressed this by predicting discrete speech tokens, but they suffer from high inference latency and exposure bias. Non-autoregressive (NAR) models such as Voicebox [12] and NaturalSpeech 3 [13] improved speed through parallel generation, yet they relied on complex pipelines: phoneme aligners, duration predictors, or factorised codec representations.

F5-TTS [1] simplifies this pipeline significantly. It is a fully non-autoregressive system based on Conditional Flow Matching [2] and a Diffusion Transformer (DiT). Instead of requiring a phoneme aligner or duration predictor, it pads the raw character sequence with filler tokens to match the mel length and lets the DiT learn alignment implicitly through a speech-infilling objective. A ConvNeXt V2 text encoder refines the character embeddings before they enter the transformer, and an inference-time Sway Sam-

pling strategy biases the ODE solver toward early timesteps, boosting intelligibility without retraining. Together, these choices yield near-perfect transcription accuracy with only 32 neural function evaluations.

Despite these gains, F5-TTS's own conclusion notes a critical gap: *"F5-TTS lacks fine-grained control of paralinguistic details, e.g. emotion."* [1]. The model can replicate *who* is speaking but cannot independently control *how* they speak - their emotional tone, energy, or prosodic style. Addressing this gap is the central motivation of this work. We use "style transfer" broadly to encompass any inference-time manipulation that shifts the acoustic characteristics of generated speech toward a reference signal - including acoustic morphing where identity and style are not fully disentangled.

We make the following contributions:

(1) **Baseline reproduction**: we replicate the sway-sampling and CFG strength ablations from the original paper, validating our pipeline.

(2) **Direct mel injection** (Extension 1): we expose a dormant 3D conditioning branch in `cfm.py` that accepts mel spectrograms directly, bypassing the vocoder, and use it to blend two reference speakers' conditioning mels at inference time.

(3) **Noise-biased style transfer** (Extension 2): four inference-time methods: (A) biasing the ODE starting noise toward a style reference's mel; (B) steering the vector field via a two-pass style guidance term; (C) switching the conditioning mel from style to identity at a scheduled ODE timestep; (D) transferring the style reference's global amplitude statistics onto the initial noise.

(4) A discussion of cross-lingual limitations (Hebrew) and a novel adversarial threat model enabled by the noise initialisation interface.

All experiments use the publicly available pretrained F5-TTS checkpoint. No fine-tuning or model modification (beyond the `cfm.py` extension) is performed. Due to the exploratory nature of this project, all experiments use a small evaluation set (three sentences, one speaker pair).

2 Background

2.1 Conditional Flow Matching

F5-TTS generates mel spectrograms by solving an ODE:

$$\frac{d\psi_t(x_0)}{dt} = v_t(\psi_t(x_0),\, c),$$
$$\psi_0(x_0) = x_0, \quad \psi_1(x_0) = x_1, \tag{1}$$

The ODE transforms a random noise sample $x_0 \sim \mathcal{N}(0, I)$ into a clean mel spectrogram x_1. At every timestep t, the learned vector field v_t tells the solver which direction to move, given the current state and a conditioning signal $c = (x_{\text{ref}},\, z_{\text{ref}\cdot\text{gen}})$ - the reference speaker's mel concatenated with the target character sequence along the time axis. The model is trained with the Conditional Flow Matching objective [2]: rather than learning to reverse a many-step noising process as in DDPM [3], it learns the velocity v_t along a straight path that connects each noise sample to its corresponding clean mel, making both training and inference simpler.

At inference time, classifier-free guidance (CFG) [5] strengthens the influence of the conditioning signal. The model is evaluated twice per ODE step: once with the full conditioning c (reference mel + text) and once with the conditioning dropped ($\varnothing$). The difference between these two outputs isolates the effect of the conditioning; scaling it by a factor λ controls how strongly the output follows the reference speaker:

$$v_{\text{guided}} = v_t(x, \varnothing) + \lambda\big[v_t(x, c) - v_t(x, \varnothing)\big]. \tag{2}$$

When $\lambda = 1$ the conditioning acts at its natural strength; $\lambda > 1$ amplifies it, improving speaker fidelity at the risk of reduced intelligibility (as confirmed by the ablation in Section 3).

2.2 Sway Sampling

F5-TTS replaces the uniform ODE timestep grid with a non-uniform schedule

$$t' = t + s\big(\cos\tfrac{\pi}{2}t - 1 + t\big)\,, \tag{3}$$

which, for $s < 0$, concentrates function evaluations near the noisy end ($t \approx 0$). The paper demonstrates that $s = -1$ with NFE $= 32$ minimises WER [1].

2.3 Evaluation Metrics

Three complementary metrics are used throughout: **WER** (word error rate) via Whisper large-v3-turbo [6] measures intelligibility. **SIM** (speaker similarity), computed as the cosine similarity of mean-pooled WavLM-Base-Plus [7] embeddings, measures how closely the generated audio matches a reference speaker. In single-reference experiments we write SIM-o (similarity to the sole conditioning reference); where two references are active, SIM-A denotes similarity to the identity reference and SIM-B to the style reference. **MCD** (mel-cepstral distortion, 24 MFCCs, DTW-aligned) measures acoustic distance from a reference in cepstral space.

3 Baseline Reproduction

3.1 Sway Sampling Ablation

We reproduced the sway-sampling experiment from Fig. 3 / Table 2 of the base article over $s \in \{+0.4, 0.0, -0.4, -0.8, -1.0\}$ and NFE $\in \{8, 16, 32\}$, for three test sentences (45 inferences total). Table 1 reports WER averaged across sentences for selected s-NFE combinations.

Table 1: Sway sampling ablation: mean WER (%) and SIM-o over three sentences. Best result per NFE shown in **bold**.

s	NFE	WER (%)	SIM-o
+0.4	8	87.9	0.694
+0.4	32	51.9	0.827
0.0	32	3.7	0.836
−0.4	32	7.9	0.826
−0.8	32	**0.0**	0.784
−1.0	8	**20.7**	0.830
−1.0	32	7.4	0.802

Key finding: $s = -0.8$ with NFE $= 32$ achieves 0 % WER on all three sentences. The qualitative trend matches the base article's Fig. 3 (right panel): WER decreases monotonically as s becomes more negative, and the benefit is largest in the low-NFE regime-at NFE $=$ 8, $s = -1.0$ yields a 67 % WER reduction over $s = +0.4$. Conversely, SIM-o rises as s moves toward positive values (0.784 at s=−0.8 vs. 0.836 at s=0.0). This reveals an intelligibility-fidelity trade-off inherent to sway sampling: negative s concentrates ODE evaluations near the noisy end ($t \approx 0$), where coarse structure such as word identity is resolved, but leaves fewer steps near the clean end ($t \approx 1$), where fine-grained speaker characteristics are refined. Positive s reverses this allocation, improving speaker similarity at the cost of intelligibility. Note that Fig. 3 of the base article reports results for 155 M parameters small models on Seed-TTS *test-zh* (Mandarin), whereas our sweep uses the 336 M base model on three English sentences; the trend's consistency across model scale and language strengthens the finding. Our $s = -1$, NFE $= 32$ result (WER = 7.4 %, SIM-o = 0.802) is the closest point of comparison with the base article's Table 2, which reports WER = 1.83 % and SIM-o = 0.67 on Seed-TTS *test-en* under identical hyperparameters; the gap is attributable to our much smaller evaluation set (3 vs. 1 088 utterances) and different speaker embedding model (WavLM-Base-Plus vs. WavLM-large-based). The optimal s shifts from -1 to -0.8 in our setup, likely a small-sample artefact. For all Extension 2 experiments we retain the F5-TTS default $s = -1.0$ to ensure comparability with the base article. Figure 1 illustrates the resulting timestep densities.

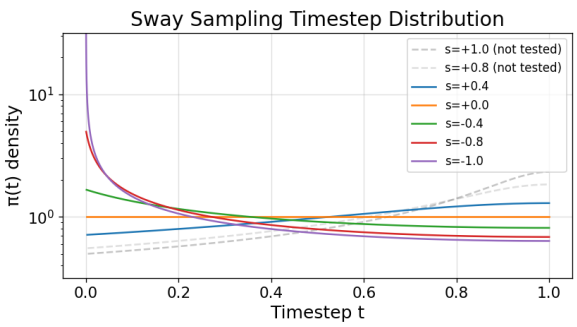


Figure 1: ODE timestep probability densities for each sway coefficient. Negative s concentrates evaluations near $t = 0$ (the noisy end); the shape matches Fig. 2 of the base article.

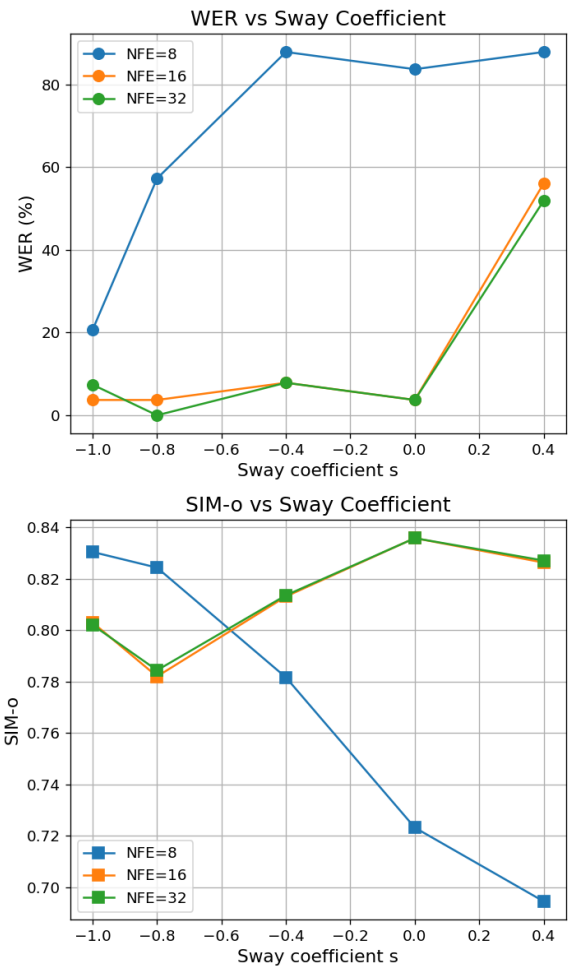


Figure 2: WER and SIM-o vs. sway coefficient for NFE$\in \{8, 16, 32\}$. The cross-over between NFE curves demonstrates that negative sway matters most when fewer ODE steps are taken.

3.2 CFG Strength Ablation

The base article adopts a default CFG strength of $\lambda = 2.0$ but does not sweep this hyperparameter. Table 2 fills that gap by evaluating $\lambda \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$ at fixed $s = -1.0$, NFE $= 32$. WER improves monotonically with stronger guidance (17.2 %→3.7 %), while SIM-o peaks at $\lambda = 1.0$-1.5 and then falls (0.835→0.774), revealing a clear intelligibility-similarity trade-off. The default $\lambda = 2.0$ (WER = 7.4 %, SIM-o = 0.802) sits at the elbow of this trade-off, providing empirical justification for the base article's choice.

Table 2: CFG strength ablation: mean WER (%) and SIM-o.

λ	WER (%)	SIM-o
1.0	17.2	0.835
1.5	11.1	0.821
2.0	7.4	0.802
2.5	3.7	0.756
3.0	3.7	0.774

4 Extension 1: Direct Mel Conditioning Interpolation

4.1 Motivation and Architecture Insight

F5-TTS conditions the DiT on a reference mel by concatenating it (feature-axis) with the noisy input and projecting the result. Inspection of `cfm.py` (lines 106-109) reveals a dormant branch:

`if cond.ndim == 2:` convert waveform → mel

When a 3D tensor `cond` $\in \mathbb{R}^{B\times T\times 100}$ is passed, the conversion is skipped entirely - the caller supplies the mel directly. This interface, present in the published codebase but unexploited in the official CLI, enables us to inject *arbitrary* mel tensors as conditioning with zero architectural change.

4.2 Mel Blending

Given identity reference A and style reference B with mels $\mathbf{m}_A, \mathbf{m}_B \in \mathbb{R}^{T\times 100}$ (computed via the model's own `MelSpec` module to remain

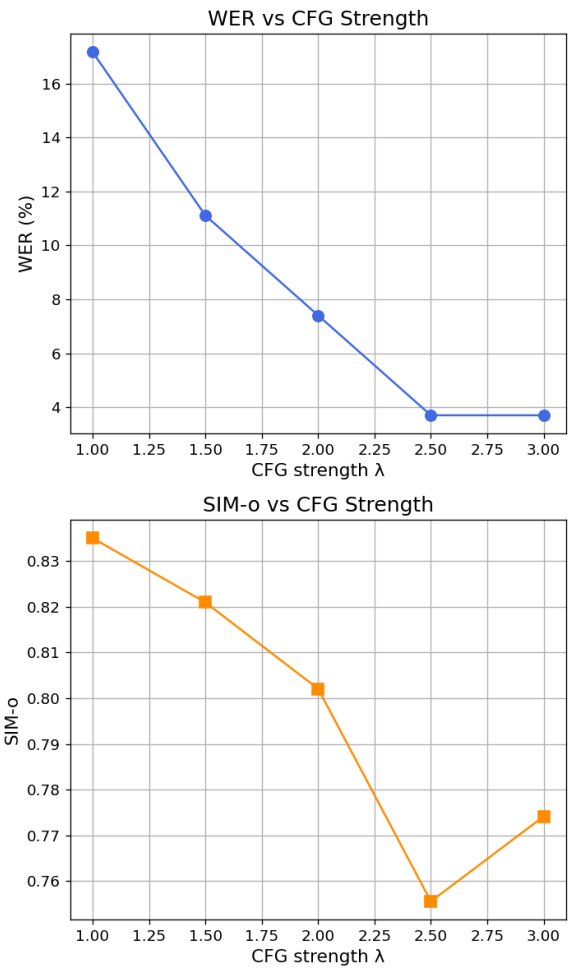


Figure 3: CFG strength ablation: WER decreases and SIM-o peaks at $\lambda \approx 1.0$-1.5 before falling, confirming $\lambda = 2.0$ as a good operating point.

in-distribution), we form

$$\mathbf{m}(\alpha) = (1-\alpha)\,\mathbf{m}_A + \alpha\,\mathbf{m}_B, \quad \alpha \in [0,1]. \quad (4)$$

Both mels are extracted using the same filterbank configuration that the model was trained on (24 kHz, 100 channels, hop = 256, Slaney norm). This means the blended tensor has the same scale and frequency layout the transformer expects, so it remains in-distribution - even though the model never saw a blended conditioning signal during training.

Why not blend waveforms or invert first? We considered two simpler alternatives but both fail. Mixing raw waveforms causes phase cancellation between the two signals, destroying the spectral structure the model relies on. Blending in mel space and then converting back to audio via the Vocos vocoder also fails: the vocoder has never seen a blended mel during training, so at intermediate weights ($\alpha \in (0.1, 0.9)$) it produces short, garbled clips (100 % WER in our early experiments). Our solution - injecting the blended mel directly into the transformer's conditioning input - avoids the vocoder entirely and keeps the signal in the representation space the model already understands.

Important limitation. A mel spectrogram encodes identity (vocal tract shape, voice quality), prosody (pitch, rate, energy), and emotional style *jointly*. Linear blending therefore produces an acoustic morph between the two speakers - not "A's voice with B's emotion." True identity-preserving emotion transfer would require a disentangled representation (separate speaker and emotion encoders) that F5-TTS does not possess.

4.3 Results

Identity reference A: canonical English male demo (`basic_ref_en.wav`). Style reference B: Mandarin animated character (`basic_ref_zh.wav`). Target text (same for all runs): *"I don't really care what you call me. I've been a silent spectator, watching species evolve, empires rise and fall."*

The metric trajectories in Fig. 4 confirm a smooth acoustic gradient: SIM-o decreases (A's identity is diluted) and MCD increases (output moves away from A) as α grows. Duration also

Table 3: Direct mel injection results. SIM-o is cosine similarity vs. reference A; MCD is mel-cepstral distortion vs. A.

α	Dur (s)	WER (%)	SIM-o	MCD
0.00	12.3	0.0	0.893	726
0.25	12.3	0.0	0.896	707
0.50	12.3	15.0	0.833	870
0.75	13.3	0.0	0.812	832
1.00	13.3	0.0	0.811	919
zero-shot B	16.9	0.0	0.840	846

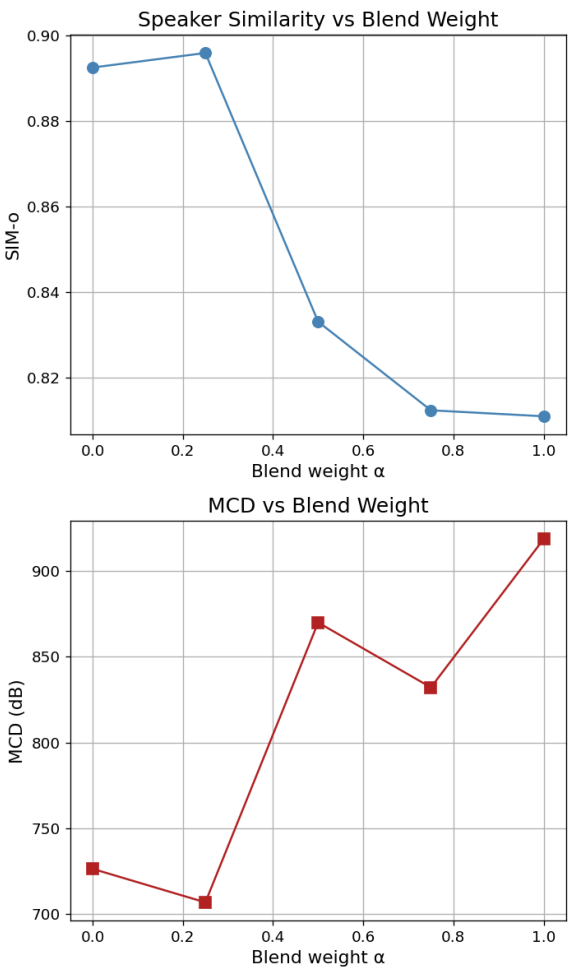


Figure 4: SIM-o and MCD as a function of blend weight α. Both metrics shift monotonically, confirming a smooth acoustic gradient between the two reference speakers.

shifts from A's pace (12.3 s) toward B's (13.3 s) at $\alpha \geq 0.75$, reflecting B's conditioning mel's influence on prosodic rhythm.

The $\alpha = 0.5$ artefact (WER = 15 %) is noteworthy: reference A's transcript contains "Some call me nature..."; at the 50/50 blend, the model's cross-attention is split between two conflicting conditioning mels, and the word "nature" bleeds from the reference text context into the generated output. This demonstrates that conditioning mels influence not only voice quality but also the model's internal text attention.

5 Extension 2: Noise-Biased Style Transfer

5.1 Overview

Extension 1 manipulates *what the model listens to* (the conditioning). Extension 2 takes a fundamentally different approach: leave the conditioning signal pure (always speaker A) and instead bias the ODE *starting point* x_0 toward speaker B's acoustic structure.

This mirrors SDEdit [4]: in image editing, SDEdit corrupts a target image to a chosen noise level then denoises it under a new prompt. Here, we corrupt/blend speaker B's mel with Gaussian noise:

$$x_0 = (1-\alpha)\,\varepsilon + \alpha\,\mathbf{m}_B, \quad \varepsilon \sim \mathcal{N}(0, I), \tag{5}$$

and integrate the ODE from $t_{\text{start}} = \alpha$ conditioned throughout on $\mathbf{m}_A$. Four methodological variants are studied.

5.2 Method A - SDEdit Initialisation

Formulation. Eq. (5) with identity conditioning throughout.

Geometric interpretation. Biasing x_0 toward $\mathbf{m}_B$ places the trajectory's starting point in a neighbourhood of B's mel in noise space; the identity conditioning then acts as a gravitational pull reorienting the trajectory toward A's data manifold. The trade-off is controlled by α: at small α the effect is subtle; at large α the model must "undo" B's structure with only $1-\alpha$ integration time, degrading intelligibility.

Results. We performed a fine-grained grid search over $\alpha \in \{0.00, 0.05, \ldots, 0.70\}$ (step 0.05)

and $s \in \{-1.0, -0.9, \ldots, +1.0\}$ (step 0.1), yielding 315 inferences. The WER-SIM-B trade-off is illustrated in Fig. 6. The optimal operating point (WER = 0 %, maximum SIM-B) is at $\alpha = 0.05$, $s = -0.9$: WER = 0 %, SIM-A = 0.874, SIM-B = 0.783, MCD = 721.9 (vs. baseline SIM-B = 0.771 at $\alpha = 0$). Style leakage from B is modest but measurable; beyond $\alpha > 0.2$, WER degrades rapidly (40-100 %) as B's temporal structure dominates the initialisation.

Critically, positive sway ($s > 0$) degrades intelligibility even at $\alpha = 0$ (WER exceeds 30 % at $s \geq +0.1$, reaching 100 % at $s \geq +0.6$), confirming that concentrating ODE evaluations near $t = 1$ (the clean end) is harmful regardless of initialisation.

5.3 Method B - Style Guidance (2-Pass ODE)

Formulation. At each ODE step the transformer is evaluated twice - once conditioned on the identity reference $\mathbf{m}_A$ and once on the style reference $\mathbf{m}_B$ - to produce two velocity proposals v_A and v_B. These are blended into a single update direction:

$$v_{\text{style}} = v_A + g\,(v_B - v_A), \tag{6}$$

where $v_A = v_{\text{guided}}(x, \mathbf{m}_A, t)$, $v_B = v_{\text{guided}}(x, \mathbf{m}_B, t)$, and g is the style guidance scale. The resulting v_{style} replaces the usual velocity v_t in the ODE solver (Eq. 1): at every integration step the solver advances x in the direction of v_{style} rather than the single-speaker velocity, steering the trajectory toward B's acoustic characteristics while remaining anchored to A's identity. This is structurally analogous to classifier-free guidance (Eq. 2), but the roles differ: λ in CFG scales the effect of conditioning vs. no conditioning, whereas g here scales the shift from one speaker's conditioning toward another's. We use a distinct symbol to reflect this difference. Because computing v_{style} requires two transformer forward passes per step (one for each conditioning mel), the computational cost doubles relative to standard inference.

Results. We swept $g \in \{0.25, 0.5, 0.75, 1.0, 1.5\}$. At $g = 0.5$,

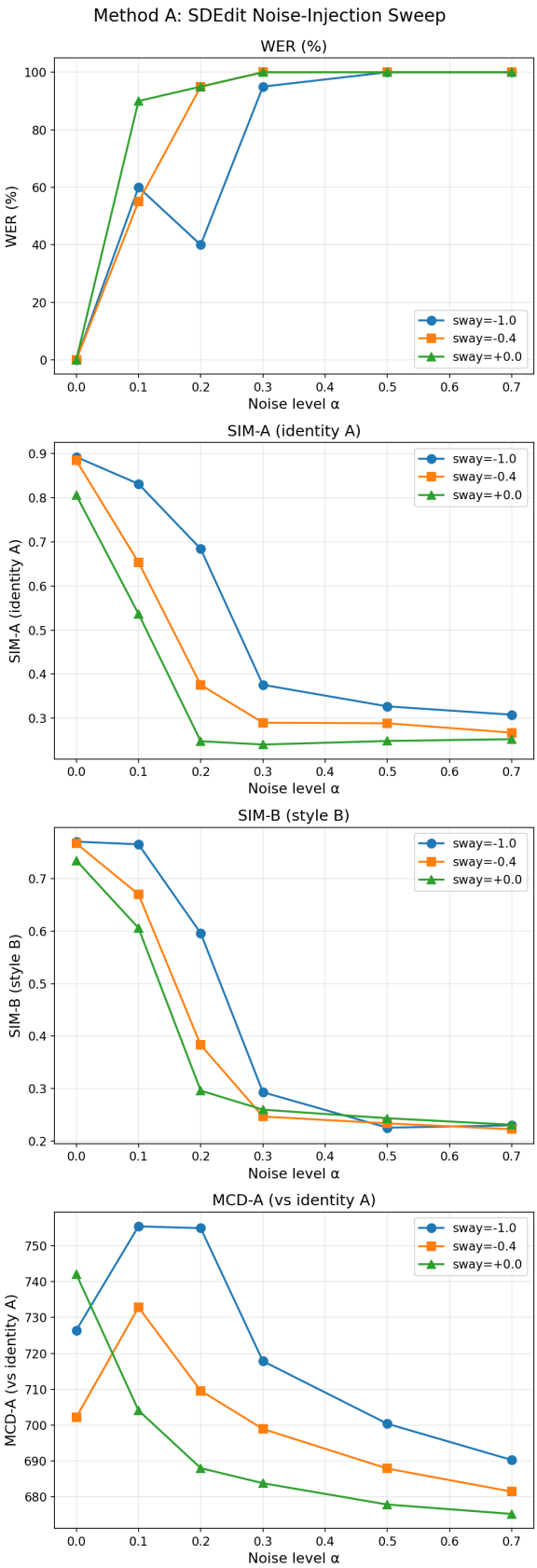
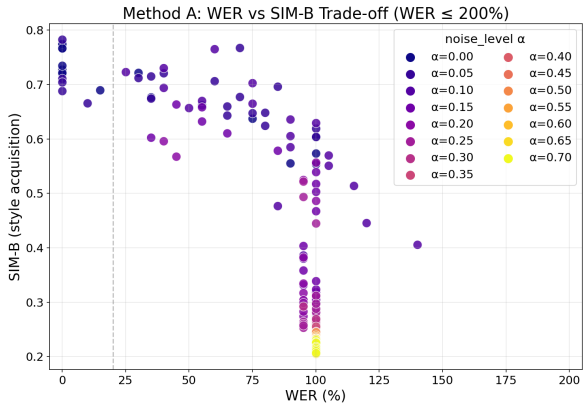
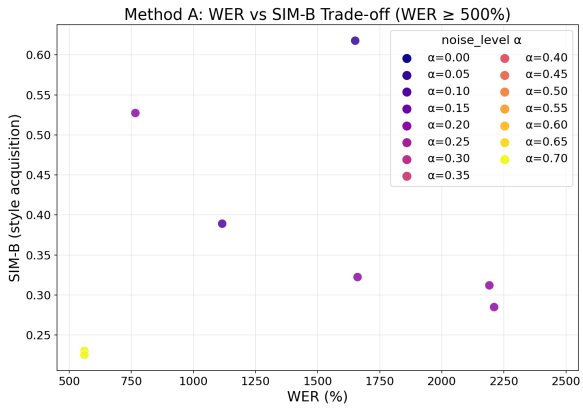


Figure 5: Method A four-metric sweep: WER, SIM-A, SIM-B, and MCD-A as functions of α for $s \in \{-1.0, -0.4, 0.0\}$. WER begins rising sharply at $\alpha > 0.1$ for all sway values.



(a) Low-WER region (WER $\leq$ 200%). The Pareto-optimal region (low WER, high SIM-B) is at small positive α with negative sway.



(b) High-WER region (WER $\geq$ 500%). These points correspond to large α values where B's temporal structure dominates the initialisation, producing unintelligible speech.

Figure 6: Method A: WER vs. SIM-B scatter for all (α, s) pairs in the fine-grained sweep. Each point is one inference; colour encodes α. WER above 100% symbolizes words generation that are not present at the target's text input.

$s = -1.0$: WER = 0 %, SIM-A = 0.843, SIM-B = **0.858**, MCD = 809. SIM-B rises by +0.087 above baseline with zero intelligibility loss - the best style transfer of all four methods. At $g \geq 1.0$ intelligibility collapses (WER $\geq$ 70 %) as the vector field is extrapolated beyond the style manifold.

5.4 Methods C & D - Negative Results

Two additional variants were tested but failed to produce intelligible style transfer at any non-trivial parameter setting.

Method C (Scheduled Conditioning) modifies the conditioning mel that is fed to the transformer's vector field v_t (Eq. 1) at each ODE step, switching it from the style reference to the identity reference at a threshold timestep t^*:

$$c(t) = \begin{cases} \mathbf{m}_B & t < t^* \\ \mathbf{m}_A & t \geq t^* \end{cases} \tag{7}$$

The hope is that conditioning on $\mathbf{m}_B$ during early steps imprints B's stylistic characteristics, while switching to $\mathbf{m}_A$ for the remaining steps restores A's identity. However, as discussed in Section 3, early ODE steps ($t \approx 0$) primarily resolve coarse structure such as word identity, not fine-grained speaker traits - so conditioning on $\mathbf{m}_B$ early actually imprints B's identity wholesale, leaving too little integration time for $\mathbf{m}_A$ to recover. All $t^* > 0$ cause severe WER degradation (20-70 %), and informal listening confirms the model adopts B's voice wholesale rather than borrowing stylistic attributes-identity *replacement*, not style transfer. The sharp conditioning discontinuity overwhelms the Euler integrator; a smooth cosine schedule may mitigate this, but is left as future work.

Method D (Noise Statistics Transfer) rescales x_0 to match $\mathbf{m}_B$'s global mean and variance:

$$x_0 = \frac{\varepsilon}{\|\varepsilon\|_\sigma} \cdot \sigma_{\text{target}} + \alpha\, \mu_B, \tag{8}$$

where $\sigma_{\text{target}} = \alpha\, \text{std}(\mathbf{m}_B) + (1 - \alpha)\, \text{std}(\varepsilon)$ and $\mu_B = \text{mean}(\mathbf{m}_B)$. At all non-zero α, WER = 100 %. This strong negative result reveals that the model is sensitive to the *variance* of x_0, not only its spatial structure: rescaling

destroys the implicit amplitude calibration the ODE integration relies on.

5.5 Cross-Method Comparison

Table 4 summarises the best intelligibility-preserving (WER = 0 %) result from each method at $s = -1.0$. Figure 7 compares the WER-SIM-B scatter across all methods. Note that Methods C and D achieve WER = 0 % only at their trivial settings (t^*=0 and α=0), which reduce to the baseline.

Table 4: Best WER = 0 % result per method at $s = -1$. Baseline: $\alpha = 0$, $g = 0$, $t^* = 0$.

Method	WER	SIM-A	SIM-B	MCD
Baseline	0 %	0.893	0.771	726
A ($\alpha = 0.05$)	0 %	0.874	0.783	722
B ($g = 0.5$)	0 %	0.843	**0.858**	809
C ($t^* = 0$)	0 %	0.893	0.771	726
D ($\alpha = 0$)	0 %	0.893	0.771	726
Ext. 1 ($\alpha = 0.25$)	0 %	SIM-o = 0.896		707

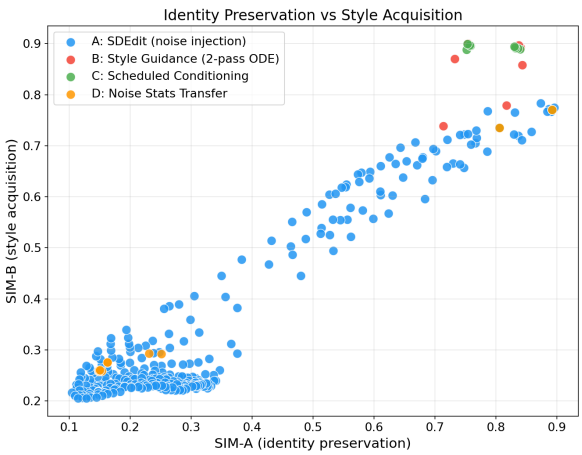


Figure 7: Identity preservation (SIM-A) vs. style acquisition (SIM-B) for all inferences across Methods A-D. Each point is one inference; the ideal operating region is the upper-right corner (high SIM-A and high SIM-B). Method B achieves the highest SIM-B at WER = 0 %; Methods C and D achieve high SIM-B only at the cost of unintelligible output.

The comparison reveals a clear hierarchy. Method B is the most expressive (ΔSIM-B = +0.087) but requires 2× compute. Method A provides modest but clean style transfer (ΔSIM-B = +0.012) at no additional cost. Methods C and D yield high SIM-B figures only in configurations where speech is unintelligible - hollow wins that do not represent functional style transfer. Extension 1's mel blending is complementary: it achieves the lowest MCD (707 at $\alpha = 0.25$) and the highest SIM-o, but at the cost of joint identity-style blending rather than clean style separation.

5.6 Informal Listening Impressions

Two of the authors independently listened to selected samples from each experiment. Key observations:

- **Baseline Reproduction** (Section 3): intelligible but with a slightly robotic, "phone-call" quality (naturalness 3/5). A noticeable accent shift toward British English was observed relative to the American-accented reference.
- **Ext. 1,** α=0: near-perfect identity fidelity and naturalness (both 5/5); indistinguishable from a standard F5-TTS zero-shot output.
- **Ext. 1,** α=0.5: a moderate acoustic morph is perceptible (3/5), and speech remains fully intelligible (5/5). Brief gibberish is audible at the utterance onset, consistent with the 15 % WER; a faint buzzing artefact persists in the background.
- **Ext. 1,** α=1.0: the voice shifts almost entirely to reference B, confirming that identity is not preserved at full blend weight (style shift 3/5, naturalness 4/5).
- **Method A** (α=0.05, s=−0.9): only a subtle difference from the pure baseline is audible (2/5); the output retains strong American phonetic features (e.g. rhotic /r/) with only a hint of reference B's characteristics. Fully intelligible (5/5).
- **Method B** (g=0.5, s=−1): the most perceptible style transfer - the voice retains speaker A's identity but carries reference B's energy contour and prosodic accent (style acquisition 3-3.5/5). Fully intelligible (5/5); minor electronic noise at the utterance onset.
- **Method C** (t^*=0.25): at sway= 0 the output is partially intelligible (3/5) with noticeable electronic noise (naturalness 2/5); at sway= −1 intelligibility degrades further. In

both cases the voice sounds like speaker B rather than A (style acquisition 4/5), confirming that scheduled conditioning produces identity *replacement* rather than style transfer - the model adopts B's voice wholesale instead of borrowing only stylistic attributes.

- **Method D**: even at α=0 (no style injection), the output degrades into robotic, distorted speech with heavy electrical background noise partway through the utterance. At any $\alpha > 0$ the output is pure noise. This confirms the quantitative finding: rescaling noise variance disrupts the ODE integration so severely that no usable speech is produced.

These informal impressions are consistent with the quantitative metrics: Method B produces the largest perceptual style shift, matching its highest ΔSIM-B, while Method A's modest SIM-B gain corresponds to a barely perceptible auditory difference.

6 Discussion

6.1 Cross-Lingual Limitation: Hebrew

F5-TTS cannot be applied to Hebrew without substantial retraining. Its character-level tokeniser (2546 tokens) covers only English and Chinese; Hebrew characters produce out-of-vocabulary tokens, rendering text conditioning meaningless. Even with vocabulary extension, Hebrew phonology includes sounds absent from both training languages (pharyngeal fricatives, the approximant *ayin*, the affricate *tsadi*), so the learned phoneme - acoustic mapping would not generalise. Our Extensions 1 and 2 operate at the acoustic level and presuppose functional text conditioning - they cannot compensate for a missing tokeniser. Meaningful Hebrew TTS would require tokeniser extension, a large Hebrew speech corpus, and compute-intensive retraining of the 335 M-parameter model. All in all, based on preliminary tests run in the scope of this work, using Hebrew text either as input or zero shot yields a gibberish voice (no-word spoken) of the target.

6.2 A Theoretical Security Concern

Extension 2 demonstrates that x_0 is a controllable interface. This opens a dual-use concern: by analogy with adversarial examples in discriminative models [8, 9], a crafted x_0 could steer the generated speech away from what the text and conditioning specify - e.g. injecting unintended content or embedding covert acoustic commands for downstream voice-activated systems [10]. Method A already shows that structured x_0 systematically shifts the output; a gradient-optimised variant could target a specific adversary-chosen mel while appearing statistically Gaussian. Mitigations include server-side sampling of x_0 from a cryptographically secure RNG, statistical monitoring of submitted noise vectors, and output-side safety filters. We emphasise that this is a theoretical observation; no attack is attempted.

7 Conclusion and Future Work

We have demonstrated two families of inference-time acoustic style transfer for F5-TTS, without modifying model weights. Extension 1 (direct mel injection) provides a smooth acoustic gradient between reference speakers via a dormant but architecturally principled 3D conditioning branch; it is the simplest intervention and yields the lowest MCD, though it blends identity and style jointly. Extension 2 (noise-biased ODE initialisation) cleanly separates identity conditioning from style injection; Method B (2-pass style guidance) achieves the largest measurable style transfer (ΔSIM-B = +0.087) at the cost of doubled inference time, while Method A provides a cheaper alternative with modest but zero-WER style leakage.

The comparison of Methods A-D illuminates the geometry of the flow-matching ODE: the model is highly sensitive to the *variance* of x_0 (Method D fails immediately), tolerates frame-level biasing at small α (Method A), and might follow a smoothly changing conditioning field provided it is not switched abruptly (Methods B vs. C).

Several directions remain for future work. **Disentangled style transfer:** true emotion-only control requires a model with separate speaker and emotion encoders; fine-tuning F5-TTS with

an auxiliary emotion classifier loss is one route. **Hebrew adaptation:** as argued in Section 6.1, meaningful Hebrew TTS requires tokeniser extension and retraining on a multilingual corpus-a worthwhile long-term goal as Hebrew speech datasets grow. **Smooth scheduled conditioning:** replacing the step function of Method C with a cosine interpolation schedule may recover intelligibility while retaining the computational efficiency advantage. **Adversarial robustness:** the security threat described in Section 6.2 deserves some empirical study; gradient-based optimisation of x_0 against a target mel would quantify the practical risk.

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